

Shagun Shaw

Backend Developer

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SUMMARY

Backend developer with hands-on experience building production-grade systems on Node.js/Express, spanning relational and in-memory data layers (PostgreSQL with Drizzle ORM, Redis for caching and atomic state), background job processing (Bull Queue), and real-time architectures (Socket.io, JWT-based auth built from scratch). Comfortable owning a service end-to-end: schema design, concurrency-safe backend logic, and deployment on AWS (EC2, Nginx, PM2, CI/CD via GitHub Actions) through to production monitoring.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, C++, SQL
Backend	Node.js, Express.js, RESTful APIs, Socket.io, JWT Auth, RBAC Middleware, Zod, Bull Queue, Nodemailer
Databases & ORM	PostgreSQL, Drizzle ORM, Redis, MongoDB, MySQL
Cloud & DevOps	AWS EC2, AWS S3, Nginx (reverse proxy, SSL/TLS), PM2, Docker, GitHub Actions (CI/CD)
Monitoring & Tools	Winston, Axiom, Git, GitHub, Postman, VS Code
Other	LLM-based App Development, RAG

PROJECTS

Tech Hunt — Real-Time Treasure Hunt Management System | *Node.js, Express.js, TypeScript, PostgreSQL, Drizzle ORM, Redis, Bull Queue, Docker* [GitHub](#)

- Engineered a full-stack real-time treasure hunt platform handling multiple concurrent participants with role-based access control (Participant, Admin, Super Admin) and Google OAuth 2.0 built from scratch, without Passport.js.
- Architected a Redis-backed group formation system with TTL-based auto-expiry, atomic INCR for race-condition-free round-robin theme distribution, and single-shot PostgreSQL transaction commits to eliminate partial-state failures.
- Implemented JWT authentication with access/refresh token architecture, session-ID-based multi-device session management, and custom RBAC middleware using higher-order functions with no third-party auth libraries.
- Designed a 6-level QR-based game progression engine with per-group theme and hint tracking in Redis, a level-based scoring system, and a Bull Queue-scheduled auto-end job with parallelized cleanup.
- Optimized data access using Drizzle ORM with full TypeScript inference, complex inner joins, subquery-based single-query membership verification, JSONB session storage, and atomic multi-table transactions.
- Hardened the system with Winston/Axiom logging, rate limiting on sensitive routes, CORS, Zod validation, and a Dockerized local development setup; parallelized question assignment across 6 domains using Promise.all.

QuizBlitz — Real-Time Quiz Hosting Platform | *TypeScript, Node.js, Express.js, Socket.io, MongoDB, JWT, AWS EC2, Nginx, PM2* [GitHub](#) | [Live Demo](#) (on Render/Vercel)

- Engineered Google OAuth 2.0 from scratch (no auth library), implementing a stateful JWT system with refresh token rotation, per-device session management, and secure httpOnly cookie handling.
- Built a Socket.io-powered real-time quiz engine with isolated in-memory room state and synchronized broadcasting; designed a custom scoring system with order-based points and streak bonus multipliers.
- Deployed on AWS EC2 behind an Nginx reverse proxy with HTTPS/SSL and a custom domain, managing the Node process with PM2 for automatic restarts (later migrated to Render/Vercel after the AWS free tier ended).
- Built a GitHub Actions CI/CD pipeline for automated SSH deployment to EC2 using GitHub Secrets and IAM-scoped credentials; centralized Winston, PM2, and Nginx logs into Axiom for production monitoring.

ACHIEVEMENTS

- Secured Top 5 Finalist position in HexaFalls2 2026, for the track "Best Use of Gemini API", recognized for innovation and technical execution.
- Core Team Member, GDG on Campus TIU (DSA domain) — DSA Lead conducting sessions and mentoring peers in data structures and algorithms.

EDUCATION

Techno India University, Kolkata <i>B.Tech, Computer Science and Engineering</i>	Aug 2023 – June 2027 GPA: 8.98
Rajasthan Vidya Mandir <i>Higher Secondary (Science)</i>	Apr 2021 – Mar 2023 92.7%